Mixology From Home

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1. Problem Definition

The system addresses the challenge of managing cocktail recipes and ingredient inventories in a structured, scalable, and reliable manner. Existing approaches rely on fragmented online searches, manual recipe books, or disconnected mobile applications, which leads to issues such as time-consuming research, difficulty tracking available ingredients, inability to find drinks based on current inventory, and lack of personalized recommendations. These limitations motivate the need for a centralized, database-driven solution that accurately models real-world entities and their interactions.

Current systems fail to connect users' ingredient inventories with available recipes in real-time, forcing users to manually cross-reference what they own against what different recipes require. Additionally, there is no efficient way to discover new drinks or receive personalized recommendations based on past preferences and ingredient availability. The proposed system aims to resolve these challenges by defining clear entities (such as Users, Drinks, Ingredients, and Favorites), establishing meaningful relationships (such as which ingredients are required for which drinks, which drinks users have favorited, and what drinks users have made), and implementing constraints that reflect real operational workflows. This database-driven approach will enable efficient searching, filtering, personalized recommendations, and intelligent shopping list generation.

2. System Users and Privileges

List all user categories that interact with the system and clearly specify what each user is allowed to do. This section should be precise and role-based (think permissions, not UI actions).

User Type 1: Casual Drinker
Privileges:
- Create and manage personal user profile
- Search for drinks based on available ingredients
- Add/remove ingredients from personal inventory
- Save drinks to favorites list
- Track drinks made and personal ratings
- Receive personalized drink recommendations

User Type 2: Event Host
Privileges:
- All Casual Drinker privileges
- Calculate ingredient quantities for multiple servings
- Generate shopping lists for missing ingredients
- Manage multiple shopping lists

User Type 3: Professional Mixologist
Privileges:
- All Casual Drinker and Event Host privileges
- Submit new drink recipes to the database
- Edit and update existing recipes (own submissions)
- Add professional tips and techniques to recipes

3. Entities and Their Descriptions

Provide a complete list of database entities, each with a clear, real-world explanation of what it represents. Do NOT list attributes here. Attributes belong in the ER diagram only.

Entity 1: User
Description: This entity represents individual users of the Mixology From Home (MFH) application. It stores personal information including name, email, password, phone, account status (Casual Drinker, Event Host, or Professional Mixologist), profile bio, and account creation date. The User entity is central to the system as it enables personalization and different privilege levels based on user type.

Entity 2: Ingredient
Description: This entity represents individual ingredients that can be used in drinks, such as spirits, mixers, garnishes, and other components. It stores the ingredient name, category (e.g., spirit, mixer, garnish, bitters, syrup, juice), unit of measurement, and whether it is a base spirit. Ingredients are the building blocks for all drink recipes in the system.

Entity 3: Drink
Description: This entity captures information about cocktail recipes within the system. It stores the drink name, description, preparation instructions, base spirit reference, flavor profile, glassware type, difficulty level, category (e.g., cocktail, mocktail, shot, punch), the creator who submitted it, and timestamps for creation and modification. Each drink record represents a unique recipe that users can search for and favorite.

Entity 4: Drink_Ingredient
Description: This associative entity links drinks to their required ingredients. It stores the specific quantity and unit needed for each ingredient in a particular recipe, along with optional preparation notes (e.g., muddled, shaken) and ingredient order for display purposes. This relationship entity is essential for determining recipe completeness and generating shopping lists.

Entity 5: User_Inventory
Description: This associative entity tracks which ingredients each user currently has at home. It represents the personal inventory of a user and stores the quantity available and unit of measurement. This enables the system to recommend drinks based on what ingredients the user actually owns.

Entity 6: Favorite
Description: This associative entity represents the many-to-many relationship between users and drinks they have saved as favorites. It stores the timestamp when a drink was added to favorites, allowing users to quickly access their preferred recipes.

Entity 7: Shopping_List
Description: This entity represents shopping lists created by users for purchasing ingredients. It stores the list name, creation date, and status (active, completed, or archived). Users can have multiple shopping lists for different occasions or planning purposes.

Entity 8: Shopping_List_Item

Description: This associative entity links shopping lists to the ingredients they contain. It stores the quantity needed, unit of measurement, whether the item has been purchased, and optional notes. This facilitates ingredient management and purchase tracking.

Entity 9: User_Drink_History
Description: This entity tracks drinks that users have actually made. It stores the date the drink was made, an optional personal rating (1-5) reflecting the user's success with the recipe, and personal notes about modifications or observations. This data enables personalized recommendations based on past behavior.

4. Relationships and Their Descriptions

List and describe how entities are connected. Focus on semantic meaning, not cardinality yet (that comes in the ER diagram). Relationships should clearly justify why they exist in the real world.

Relationship 1: User - manages - User_Inventory
Description: This relationship represents users managing their personal ingredient inventory. It exists because users need to track what ingredients they have at home and enables the drink recommendation functionality based on available ingredients.

Relationship 2: Ingredient - stored_in - User_Inventory
Description: This relationship allows the system to track which ingredients exist in user inventories by linking the master ingredient list with individual user stocks.

Relationship 3: Drink - requires - Drink_Ingredient
Description: This relationship represents the recipe composition of each drink. It exists because every drink must have at least one ingredient and enables recipe display and shopping list generation.

Relationship 4: Ingredient - used_in - Drink_Ingredient
Description: This relationship allows the system to determine which drinks use a particular ingredient by linking ingredients to their recipe appearances.

Relationship 5: Ingredient - base_spirit - Drink
Description: This relationship identifies the primary spirit for each drink. It enables filtering and searching drinks by their base spirit (e.g., show all vodka drinks).

Relationship 6: User - creates - Drink
Description: This relationship tracks which user (typically a Professional Mixologist) submitted each drink recipe. It enables attribution and allows users to edit their own submissions.

Relationship 7: User - saves - Favorite
Description: This relationship represents users bookmarking drinks they want to save for quick access. It enables the favorites list functionality.

Relationship 8: Drink - bookmarked_as - Favorite

Description: This relationship allows the system to track which drinks have been favorited by linking drinks to favorite records.

Relationship 9: User - creates - Shopping_List
Description: This relationship represents users creating shopping lists for ingredient purchases. It enables the shopping list management functionality for Event Hosts.

Relationship 10: Shopping_List - contains - Shopping_List_Item
Description: This relationship links shopping lists to their individual items. It exists because a shopping list must contain at least one item and enables item-level tracking.

Relationship 11: Ingredient - listed_in - Shopping_List_Item
Description: This relationship allows the system to identify which ingredients need to be purchased by linking the master ingredient list with shopping list items.

Relationship 12: User - logs - User_Drink_History
Description: This relationship tracks which drinks a user has made over time. It enables personalized recommendations based on drinking history and preferences.

Relationship 13: Drink - recorded_in - User_Drink_History
Description: This relationship allows the system to track how often each drink is made by linking drinks to user history records.

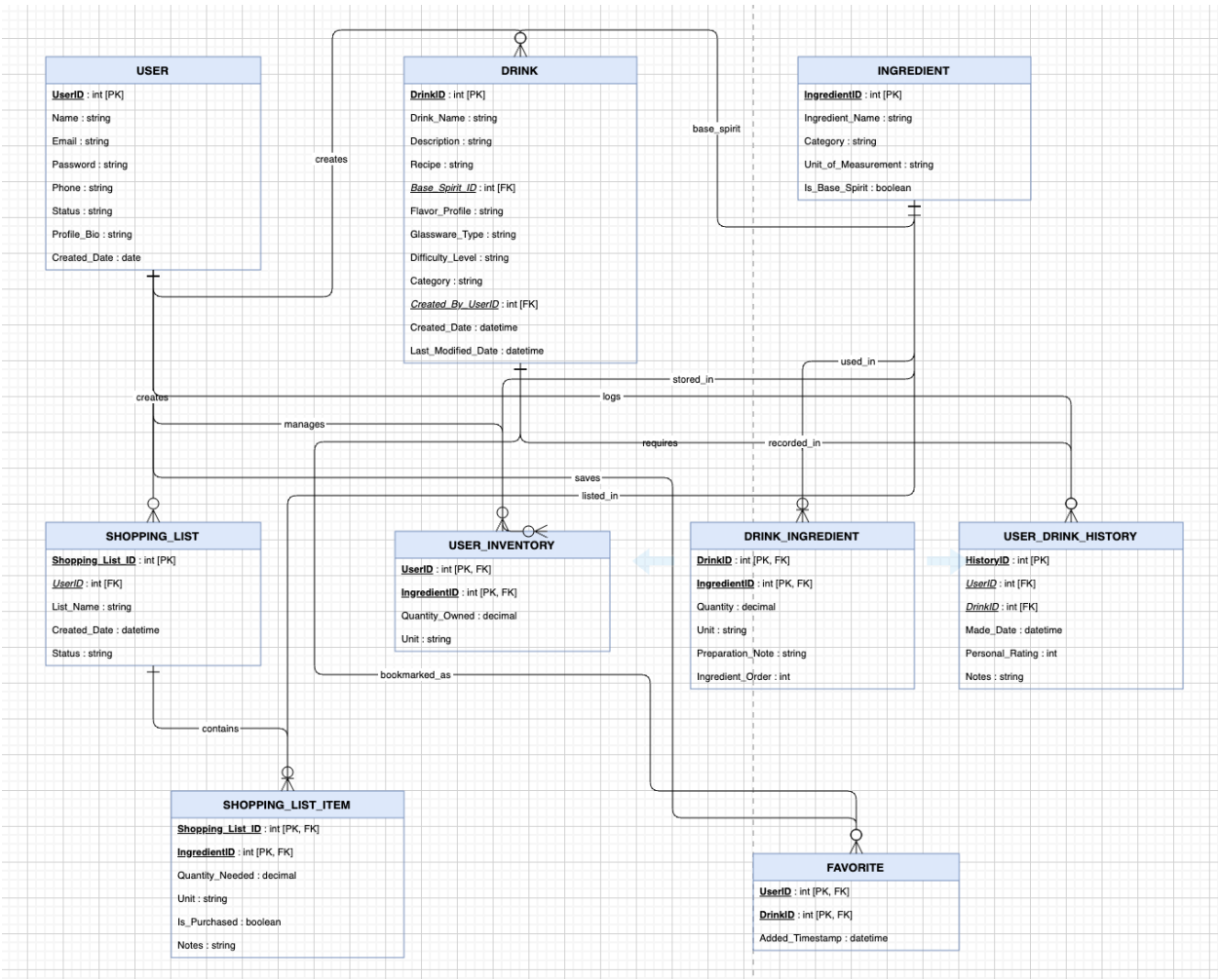
5. Entity Relationship Diagram (ERD)

Include your complete ER diagram on this page (or pages if needed).

Diagram Requirements

- Clearly labeled entities
- All attributes listed per entity
- Primary keys and foreign keys clearly identified
- All relationships shown with appropriate cardinalities
- Use Crow's Foot, Chen, or UML notation (any is acceptable)

You may create the diagram using tools such as draw.io, Lucidchart, or similar, and embed the exported image here.


USER
UserID : int [PK]
Name : string
Email : string
Password : string
Phone : string
Status : string
Profile_Bio : string
Created_Date : date
DRINK
DrinkID : int [PK]
Drink_Name : string
Description : string
Recipe : string
Base_Spirit_ID : int [FK]
Flavor_Profile : string
Glassware_Type : string
Difficulty_Level : string
Category : string
Created_By_UserID : int [FK]
Created_Date : datetime
Last_Modified_Date : datetime
INGREDIENT
IngredientID : int [PK]
Ingredient_Name : string
Category : string
Unit_of_Measurement : string
Is_Base_Spirit : boolean
creates
base_spirit
used_in
stored_in
creates
logs
manages
requires
recorded_in
saves
listed_in
SHOPPING_LIST
Shopping_List_ID : int [PK]
UserID : int [FK]
List_Name : string
Created_Date : datetime
Status : string
USER_INVENTORY
UserID : int [PK, FK]
IngredientID : int [PK, FK]
Quantity_Owned : decimal
Unit : string
DRINK_INGREDIENT
DrinkID : int [PK, FK]
IngredientID : int [PK, FK]
Quantity : decimal
Unit : string
Preparation_Note : string
Ingredient_Order : int
USER_DRINK_HISTORY
HistoryID : int [PK]
UserID : int [FK]
DrinkID : int [FK]
Made_Date : datetime
Personal_Rating : int
Notes : string
bookmarked_as
contains
SHOPPING_LIST_ITEM
Shopping_List_ID : int [PK, FK]
IngredientID : int [PK, FK]
Quantity_Needed : decimal
Unit : string
Is_Purchased : boolean
Notes : string
FAVORITE
UserID : int [PK, FK]
DrinkID : int [PK, FK]
Added_Timestamp : datetime

6. Team Roles and Contributions (Phase II)

Member 1: Griffin Witt
Role and Contributions: Every member will contribute to every part of the project

Member 2: Louie Finder
Role and Contributions: Every member will contribute to every part of the project

Member 3: Ryan Grant
Role and Contributions: Every member will contribute to every part of the project

Member 4: Timur Bashpanov
Role and Contributions: Every member will contribute to every part of the project

AI Team Member: Generative AI
Role and Contributions: Correct Spelling, Idea organization, Suggestions on certain approaches

Notes for Full Marks (Very Important)

- Problem definition must be system-focused, not abstract
- Users & privileges must be complete and non-overlapping
- Entities must align directly with Phase I real-world entities
- Relationships must be meaningful and justified
- ER diagram must be clean, readable, and accurate